

Math Exploration 1

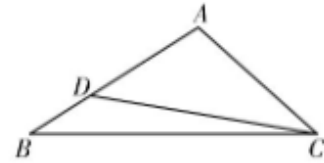
2. As shown in the figure, $\triangle ABC \sim \triangle ACD$, $AB = 10\text{cm}$, and $AC = 8\text{cm}$. The length of $\overline{AD}$ is _____.

A. 6.4cm

B. 6cm

C. 2cm

D. 4cm



Math Exploration 2

2. **SAT** Among the following, the number of conditions that can be used to deduce that $\triangle ABC$ is similar to $\triangle A'B'C'$ is _____.

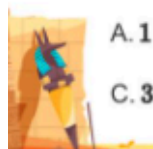
① the two angles of $\triangle ABC$ are, respectively, 58° and 70° ; the two angles of $\triangle A'B'C'$ are, respectively, 58° and 52° .

② the two side lengths of $\triangle ABC$ are, respectively, 4cm and $\frac{3}{2}\text{cm}$ and their included angle is 40° ;

the two side lengths of $\triangle A'B'C'$ are $\frac{4}{3}\text{cm}$ and $\frac{1}{2}\text{cm}$ respectively, and their included angle is 40°

③ the side lengths of $\triangle ABC$ are 5cm , 6cm , and 8cm , respectively; the side lengths of $\triangle A'B'C'$ are $\frac{5}{2}\text{cm}$, 3cm , and 4cm , respectively.

④ in $\triangle ABC$, $m\angle C = 90^\circ$, $AC = 3$, and $BC = 4$; in $\triangle A'B'C'$, $m\angle C' = 90^\circ$, $A'C' = 6$, and $B'C' = 8$.

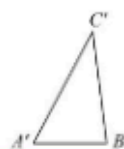


A. 1

B. 2

C. 3

D. 4



Math Exploration 3

2. In rectangle $ABCD$, $AB = 5$ and $BC = 3$. Points F and G are on $\overline{CD}$ so that $DF = 1$ and $GC = 2$. Lines AF and BG intersect at E . Find the area of $\triangle AEB$. (2003 AMC 10B Problem, Question #20)

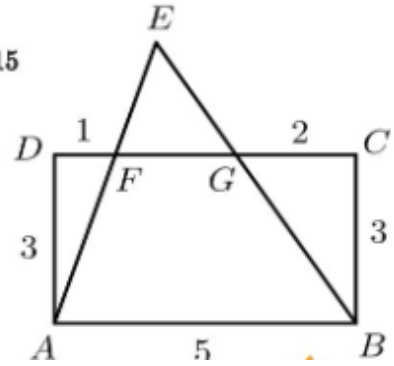
A. 10

B. $\frac{21}{2}$

C. 12

D. $\frac{25}{2}$

E. 15



Math Exploration 3

3. In $\triangle ABC$, $AB = AC = 28$ and $BC = 20$. Points D , E , and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$, respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$, respectively. What is the perimeter of parallelogram $ADEF$? (2013 AMC 12A Problem, Question #9)

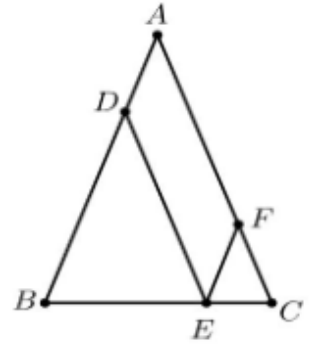
A. 48

B. 52

C. 56

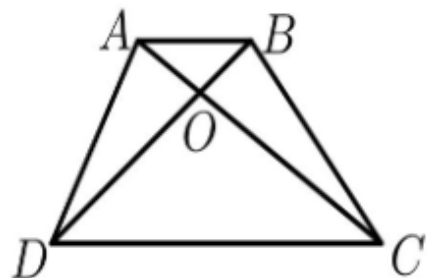
D. 60

E. 72

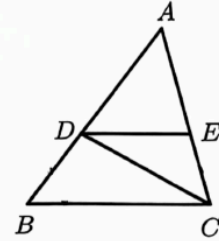


Math Exploration 4

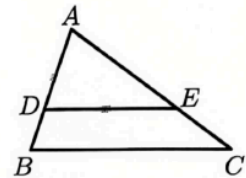
2. As shown in the figure, trapezoid $ABCD$ is divided into 4 smaller triangles by its diagonals. Given that the areas of $\triangle AOB$ and $\triangle BOC$ are 25cm^2 and 35cm^2 , respectively, then the area of the trapezoid is _____ cm^2 .



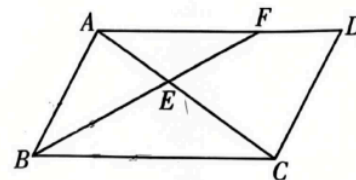
- 3 As shown in the figure, in $\triangle ABC$, point D and E are on $\overline{AB}$ and $\overline{AC}$. If $\overline{DE} \parallel \overline{BC}$, and $m\angle ACD = m\angle B$, then there are _____ sets of similar triangles in the figure.



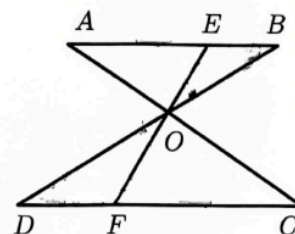
- 5 As shown in the figure, in $\triangle ABC$, point D and point E are on $\overline{AB}$ and $\overline{AC}$, $\overline{DE} \parallel \overline{BC}$. If $AD = 2$, $AB = 3$, and $DE = 4$, then $BC = \underline{\hspace{1cm}}$.



- 9 In parallelogram $ABCD$, the angle bisector of $\angle ABC$ intersects the diagonal $\overline{AC}$ at point E and intersects $\overline{AD}$ at point F . If $\frac{AB}{BC} = \frac{3}{5}$, then the value of $\frac{EF}{BF}$ is _____.



- 10 In the figure, $\overline{AB} \parallel \overline{CD}$, $\overline{AC}$ and $\overline{BD}$ intersect at point O . The line passing through point O intersects $\overline{AB}$ and $\overline{CD}$ at points E and F , respectively. The conclusion that may not necessarily be true is _____.



- A. $\frac{OA}{OC} = \frac{AB}{CD}$ B. $\frac{OA}{OD} = \frac{OB}{OC}$ C. $\frac{CD}{DF} = \frac{AB}{BE}$ D. $\frac{OE}{OF} = \frac{AB}{CD}$

Math Exploration 1

As shown in the figure, $\overline{AD} \parallel \overline{EF} \parallel \overline{BC}$, then

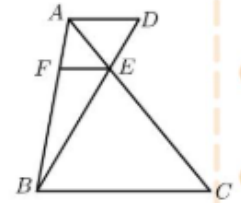
(1) $\frac{1}{AD} + \frac{1}{BC} = ?$; (2) if $AD = 3$, $BC = 5$, $AB = 12$, then calculate the length of $\overline{AF}$.

A. (1) $\frac{1}{EF}$ (2) $\frac{3}{8}$

B. (1) $\frac{1}{AF}$ (2) $\frac{9}{8}$

C. (1) $\frac{1}{AF}$ (2) $\frac{9}{2}$

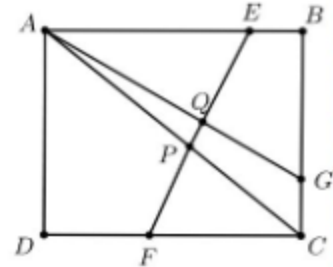
D. (1) $\frac{1}{EF}$ (2) $\frac{9}{2}$



Math Exploration 2

1. Rectangle $ABCD$ has $AB = 5$ and $BC = 4$. Point E lies on $\overline{AB}$ so that $EB = 1$, point G lies on $\overline{BC}$ so that $CG = 1$ and point F lies on $\overline{CD}$ so that $DF = 2$. Segments $\overline{AG}$ and $\overline{AC}$ intersect $\overline{EF}$ at Q and P , respectively. (Adapted from 2016 AMC 10B Problem, Question #19)

(1) What is the value of $\frac{EP}{EF}$?



Math Exploration 3

3. Rectangle $ABCD$ has $AB = 8$ and $BC = 6$. Point M is the midpoint of diagonal $\overline{AC}$, and E is on $\overline{AB}$ with $\overline{ME} \perp \overline{AC}$. What is the area of $\triangle AME$? (2009 AMC 10B Problem, Question #18)

A. $\frac{65}{8}$

B. $\frac{25}{3}$

C. 9

D. $\frac{75}{8}$

E. $\frac{85}{8}$

Math Exploration 4

2. Rectangle $ABCD$ has $AB = 3$ and $BC = 4$. Point E is the foot of the perpendicular from B to diagonal $\overline{AC}$. What is the area of $\triangle AED$? (2017 AMC 10B Problem, Question #15)

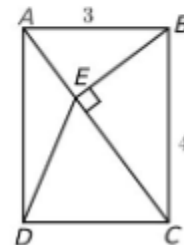
A. 1

B. $\frac{42}{25}$

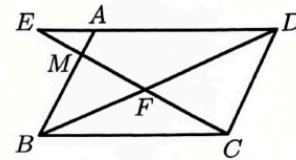
C. $\frac{28}{15}$

D. 2

E. $\frac{54}{25}$

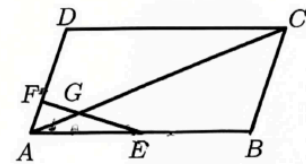


- 1 In parallelogram $ABCD$, point E is on the extension of $\overline{DA}$. Then _____ .



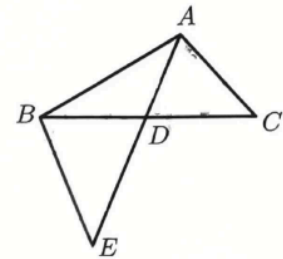
- A. $\frac{AE}{AD} = \frac{EM}{MC}$ B. $\frac{AE}{AD} = \frac{AM}{CD}$ C. $\frac{BM}{CD} = \frac{BF}{BD}$ D. $\frac{ED}{BC} = \frac{AD}{BM}$

- 2 As shown in the figure, in parallelogram $ABCD$, E is the midpoint of $\overline{AB}$, and F is a point on $\overline{AD}$. $\overline{EF}$ intersects with $\overline{AC}$ at point G . If $AF = 2\text{cm}$, $DF = 4\text{cm}$, and $AG = 3\text{cm}$, then the length of $\overline{AC}$ is _____ .



- A. 14cm B. 15cm C. 16cm D. $\frac{46}{3}\text{cm}$

- 9 As shown in the figure, in $\triangle ABC$, $\overline{AE}$ intersects with $\overline{BC}$ at point D . Draw $\overline{BE}$. If $m\angle C = m\angle E$, $AD : DE = 3 : 5$, $AE = 8$, and $BD = 4$, then the length of $\overline{DC}$ is _____.



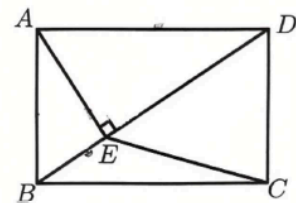
A. $\frac{15}{4}$

B. $\frac{12}{5}$

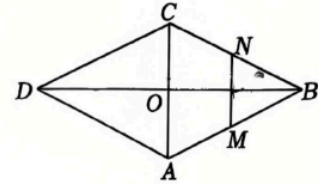
C. $\frac{20}{3}$

D. $\frac{17}{4}$

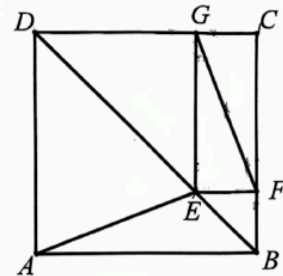
- 10 In rectangle $ABCD$, $\overline{AE} \perp \overline{BD}$ at point E . $AD = 3$, $BD = \sqrt{13}$, and draw $\overline{CE}$. Then the area of $\triangle DEC$ is _____.



- 3 As shown in the figure, O is the diagonal of rhombus $ABCD$ with AC as its intersection point. Points M and N are the midpoints of sides AB and BC , respectively, connecting MN . If $MN = 4$ and $BD = 12$, then the perimeter of rhombus $ABCD$ is _____.



- 8 As shown in the figure, point E is a point on the diagonal BD of square $ABCD$, $EF \perp BC$, $EG \perp CD$ at F , G , $GF = 3$ respectively, then $AE =$ _____.



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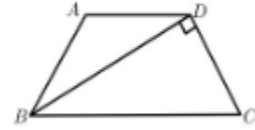
2. As shown in the figure, in the quadrilateral $ABCD$, $AD \parallel BC$, $AB = AD = DC = 1$, $BD \perp CD$, what is the area of the quadrilateral $ABCD$?

A. $\frac{\sqrt{3}}{3}$

B. $\frac{3\sqrt{3}}{2}$

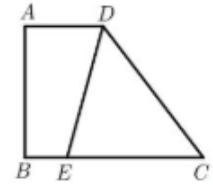
C. $\frac{3\sqrt{3}}{4}$

D. $\sqrt{3}$



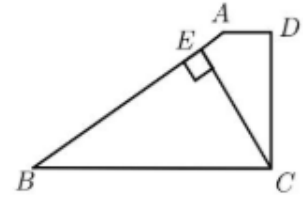
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3. As shown in the diagram, in the given trapezoid $ABCD$, $AD \parallel BC$, $\angle B = 90^\circ$, $CD = 15$, $BC = 16$, $AB = 12$. Point E is a point on side BC . Connect DE , and meet $DE = CE$. The area of trapezoid $ABCD$ is _____.



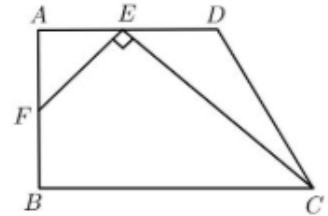
Math Exploration 1

4. As shown in the figure, in the trapezoid $ABCD$, $AD \parallel BC$, $\angle ADC = 90^\circ$, $\angle B = 30^\circ$, $CE \perp AB$, with the perpendicular foot at point E . If $AD = 1$, $AB = 2\sqrt{3}$, find the length of CE .



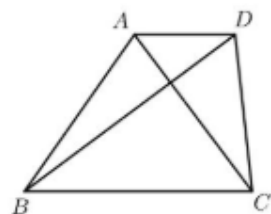
Math Exploration 1

5. As shown in the figure, in trapezoid $ABCD$, $AD \parallel BC$, $AB \perp BC$, $AB = AD = 2$, $BC = 3$, point E is the midpoint of side AD . Through point E , draw $EF \perp EC$ intersecting AB at point F . Find the length of segment AF .



Math Exploration 2

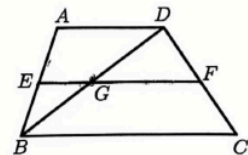
3. As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, $AD = 2$, $BC = 8$, $AC = 6$, and $BD = 8$.
The area of trapezoid $ABCD$ is _____ .



L14 HW

- 4 In the right trapezoid $ABCD$, $AD \parallel BC$, $\angle A = 90^\circ$, $AB = \frac{5\sqrt{3}}{2}$, $CD = 5$, then $\angle C =$ _____ .

- 5 As shown in the figure, the median of a trapezoid has a length of 20cm. It is divided in two parts by the diagonal, such that the difference between the lengths of two parts is 5cm. Then the length of the longer base is _____ cm.



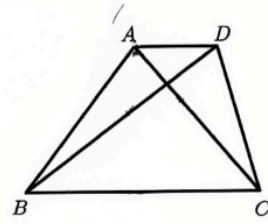
- 6 As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, $AD = 2$, $BC = 3$, $AC = 3$, and $BD = 4$. The area of $ABCD$ is _____.

A. 20

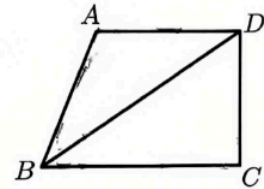
B. 12

C. 24

D. 6



- 8 As shown in the figure, in the right trapezoid $ABCD$, $AD \parallel BC$, $\angle C = 90^\circ$, $\angle A = 120^\circ$, $AD = 2$, BD bisects $\angle ABC$, then the perimeter of trapezoid $ABCD$ is _____.





It is given that the lengths of the four sides of a trapezoid are 1, 2, 3, and 4. Calculate the area of this trapezoid.

